Privacy Impact Assessment on Flock Safety Deployment in Universities

Executive Summary

System Description

Flock Safety is a private corporation that develops public security technologies, such as LPR cameras, surveillance cameras, gunshot audio detection, and drones for law enforcement, residential neighborhoods, and businesses. Our privacy impact assessment focuses on deployment in university campuses. Flock Safety is intended to boost student safety by notifying law enforcement of potential threats such as monitoring unauthorized vehicles that enter campus grounds, unpermitted individuals on campus, and locating individuals responsible for harmful activities on campus.

Key Privacy Risks Identified

Flock Safety's mass surveillance capabilities are a significant privacy risk due to the high number of devices that can process sensitive data, such as the Flock Drones, Raven Audio, and LPRs. Additionally, Flock Safety has features that search video footage based on identifiable characteristics of a person or a car, which lead to re-identification of individuals, a huge privacy risk. Finally, data retention is a serious privacy risk, especially with Flock Safety's current cybersecurity vulnerabilities, because of the amount of PII stored on the cloud.

Existing Protections

Flock's Retention Policy says that data is stored in their systems for 30 days by default in AWS cloud with AES256 encryption. Additionally, Flock is compliant with a number of security certifications which ensure Flock is able to store and manage confidential and personally identifiable information.

Proposed Mitigations

1. **Data Protections:** Incorporate technical protections such as data masking, hashing and adaptive encryption to protect data throughout its lifecycle and stricter policies to restrict access to sensitive data.
2. **Training:** Publish stricter training for customers and employees of Flock to better educate around possible privacy issues and how to address them if they arise.
3. **Policy:** Implement more signage around areas where Flock's systems are used and participate in quarterly auditing to find and eliminate possible security vulnerabilities and non-compliance.

Overall Privacy Risk Assessment

The numerous privacy risks, along with the scale at which Flock is expanding its customer base, leads us to believe Flock deployment in a university context has a very high chance to cause debilitating privacy and ethical issues in a university context causing us to designate Flock Safety a **high privacy risk.**

Recommendation

We recommend proceeding with additional oversight of the Flock Safety platform. We suggest incorporating oversights that will address how GTPD shares the Georgia Tech community data, GTPD's collaboration with Flock on preventing PII data from leaving the Georgia Tech community, training the usage of securely using Flock systems, and written policy detailing these vulnerabilities with the Flock system.

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Administrative Details

Scope of PIA

This Privacy Impact Assessment will focus on privacy concerns related to Flock Safety and its deployment in a university setting. We will mainly reference Georgia Tech as a case study, but will also touch on Flock Safety deployment on and around other universities within the PIA. The goal of this PIA is to identify potential privacy risks associated with Flock technology and how it's used, who the affected populations on and near universities are, and to mitigate risks posed by deployment.

Methodology

We analyzed the deployment of Flock Safety on university campuses using the following methodology:

1. **Primary Research -** We conducted interviews and obtained information from GTPD (Georgia Tech Police Department) personnel Chief Connolly and Captain R. McNevin, who provided first-hand accounts and details about how Flock is used at Georgia Tech, including camera counts, data access permissions, data sharing practices, and drone operations.
2. **Secondary Research** - We reviewed publicly available information from:
    a. Flock Safety
        i. Privacy policy and terms of service
        ii. Product and feature documentation and patents
    b. Independent sources
        i. Electronic Frontier Foundation (EFF) reports on ALPR
        ii. DeFlock Map
        iii. News Reports
        iv. CVEs from the National Vulnerability Database
        v. Etc.
3. **Risk Assessment -** Using our synthesis of information from collected data sources, we assigned each category a privacy risk score based on an impact-likelihood model. First, we measure on the impact on privacy, then we measure the likelihood of the privacy event occurring. The outlined risk assessment criteria is shown in Fig. 1, 2, and 3.

Assumptions

We make the following assumptions in our PIA:

Accuracy of Public Documentation: We assume Flock Safety's privacy policy and information accurately reflects how they utilize their systems.

Case Study Generalizability: We make the assumption that Georgia Tech is a strong case study to generalize to University deployments of Flock Safety.

Technical Infrastructure: We assume that any current technical information is up to date with how Flock Safety currently operates, such as any articles of the technical infrastructure.

Validity of Primary Research: We assume that the information from the chief and captain is accurate to how they are currently operating.

Leadership Dependency: We assume that the current leadership is driving the philosophy, but there is a chance for that leadership to change without the values of the previous.

Future System Expansion: We assume that there will be more investment into the Flock Ecosystem as time goes on, beyond the initial scope.

Reliance on Default Configurations: When specific information is not available, we assume default configurations are set.
Assessment Scope Boundaries: We assume the scope to be within Flock Safety and university and do not do a deep dive into the third party integrations that may be possible.
Data Flow Diagram Scope: We overview Georgia Tech's use of Flock Safety in our Data Flow Diagram to give a case study. This does not cover all universities, possible usage, or other possible third parties.

Risk Assessment Criteria

Impact Rating	Definition and Criteria
Severe	Severe harm leading to a loss of safety, systematic profiling, discrimination, or exposure of highly sensitive PII.
Moderate	Moderate harm causing distress, loss of control over personal data, or unauthorized use of data for alternative purposes.
Minor	Minor harm or annoyance involving data that is already highly public, with minimal impact on individual autonomy.

Figure 1. Impact Assessment

Likelihood Rating	Definition and Criteria
Almost Certain	The privacy event happens by default, relies purely on human policy, or the system defaults to maximum data collection.
Frequent	The privacy event is possible. Mitigations exist but can be easily bypassed, misconfigured by a user, or rely on third-party integrations.
Rarely	The privacy event is highly unlikely. Privacy-by-design principles and verifiable technical safeguards are in place, requiring deceptive intent or complex exploits to bypass.

Figure 2. Likelihood Assessment

	Impact		
	Minor	Moderate	Severe
Almost	Medium Risk	High Risk	High Risk

Likelihood				
	Certain			
	Frequent	Low Risk	Medium Risk	High Risk
	Rarely	Low Risk	Low Risk	Medium Risk

Figure 3. Final Risk Score With Impact-Likelihood Assessment

System Overview

What is Flock Safety

Flock Safety is a private corporation that develops public security technologies, such as automated license-plate readers, surveillance cameras, gunshot audio detection, and drones for law enforcement, residential neighborhoods, and businesses.

Intended Users

The intended users of Flock Safety are law enforcement agencies and organizations that seek to protect their communities with Flock technology. For Georgia Tech specifically, as detailed in our Data Flow Diagram in Fig. 12, intended users consist of Georgia Tech Police Department senior members (e.g., Lieutenant, Captain, Chief) and Flock IT department (Flock Safety CJIS-certified engineers).

Overview of Flock's Technology Ecosystem

Figure 4. Hardware Products

Product	Description	Data Collected
Falcon LPR	AI-powered License Plate Reader to scan license plates with OCR and identify vehicles based on their make, model, color, and characteristics	License plate number, plate state, plate image, full vehicle image, vehicle make, vehicle model, vehicle type, vehicle color, vehicle damage, vehicle accessories, any bystanders within frame of license plate, along with metadata including timestamp and location of each piece of data
Condor &	AI-enabled video streaming and security cameras	Continuous video of

Wing Cameras		environment, individuals, faces
Raven Audio Detection	Detects the audio of gunshots through microphone usage and ML audio classification	Detects gunshot audio and other "critical sounds" within 90 ft and provides a list of audio clips with corresponding location
Mobile Security Trailer	Station with security cameras to detect crime for quick, portable installation	Continuous video of environment, people, vehicles, faces

Figure 5. Drone Responder

Product	Description	Data Collected
Drone Alpha	The actual drone used for Drone First Responder programs	Video stream, camera sensors, bounding boxes from ML algorithms of detected objects
Drone Aerodome	The software to control the drone with mouse and keyboard, comes with a mobile app [7]	Video stream of drone, GPS location coordinates

Figure 6. Software Products

Product	Description
FlockOS®	Centralized database for devices, data, and departments
Flock Nova	Search across existing information systems [6]
Flock Mobile App	Provides real-time updates towards customers based on their infrastructure

Figure 7. Core Infrastructure

Infrastructure	Description
AI/ML Algorithms	Neural networks for vehicle detection, audio classification, natural language processing
Cloud Storage	Amazon Web Servers (AWS) for data storage and retention
Databases	SQL based databases for ML analysis with vehicle metadata

API Integrations	Developer integration with other third parties

How Flock Safety Is Deployed in University Campuses

Flock Safety brings togethers its technology ecosystem in universities to create a virtual guard around campuses. They use a philosophy of 'Detect', 'Decode', and 'Deliver'. Their hardware products have sensors to collect data, which is then processed through AI algorithms, and then synthesized in a format for the user to take action with. These devices can be placed around the outskirts of campus as shown in Figure 9 or within campus to fill any gaps within their existing security infrastructure.

The ecosystem allows you to share your data with local law enforcement partners, and give them the ability to view your data if you choose to do so. Other campuses can also share their data with each other. This creates a more interconnected network to resolve incidents [1].

University	**Scope of Deployment**	**Primary Use Purpose**	**Secondary Use Purpose**	**Data Retention Period**
Georgia Tech	15 Flock LPR Cameras 1 Flock Drone	Drones used to reduce time responding to emergencies, but mainly restricted to emergency response	Less than 1% of data is used by Flock for machine learning purposes	30 days
Virginia Tech	Specific Flock LPR Camera Data Unavailable	Bolster security measures while safeguarding individual privacy		15 days
UW-Madison	8 Flock LPR Cameras	Help law enforcement identify vehicles connected to specific criminal investigations, locate missing persons, and respond more effectively to serious safety concerns		30 days
University of Iowa	28 Flock LPR Cameras	Capture objective evidence without compromising on individual privacy and for use of retroactive search to solve crimes		30 days

Figure 8. Summary of Flock Safety deployment on Georgia Tech (R. Connolly, class interview, Feb. 23, 2026), Virginia Tech [2], UW Madison [3], and the University of Iowa [4]

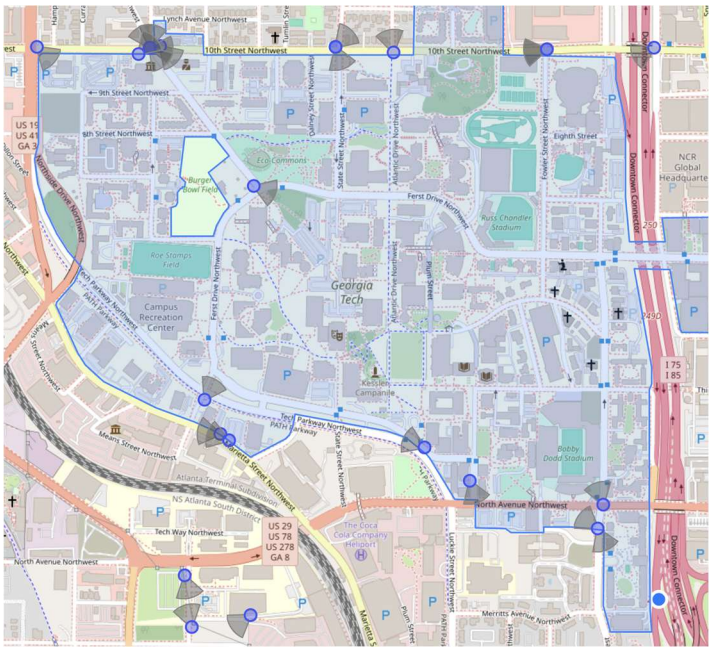


Figure 9. Example of Flock Cameras Deployed on the outskirts of the Georgia Tech's Campus from DeFlock [5]

Data Collection and Management

What Data is Collected

Question	Details
What kind of information may be collected?	• License plate images • Vehicle image • Vehicle characteristics ○ color, make • License plate number • License plate state • Date • Time • Location
What sensitive information may be collected?	• Location • Audio data (unclear scope — defined as "critical sounds") • Identifying features on a car ○ Damage, decals, bumper stickers, etc. • Patterns of travel
Who is likely captured incidentally?	• Students • Faculty • Campus visitors • Local community members

Figure 10. Summary of data collected based on Flock Safety LPR Privacy Policy [8] and from the interview with GTPD Captain R. McNevin (group interview, Feb. 25, 2026)

How Data is Stored and Shared

Question	Details
Who has access to the data, and under what conditions?	• Administrators • Authorized end-users designated by the Flock Safety customer • Flock Safety's CJIS-certified engineers • Other privileged administrators as designated Specifically at Georgia Tech: • Officers ranked Sergeant and above • Communication advisor • Detectives • Digital Investigation Team • Drones Only: Trained pilots (3 supervisors and 4 students)
Where is data stored?	Flock Safety system hosted on AWS Cloud
How long is data retained?	• Briefly on the camera, then transferred to AWS Government Cloud for a default of 30 days ○ May be adjusted on a case-by-case basis
With whom is data shared?	• Law enforcement • Government officials • Third parties ○ When legally required or in good faith belief of necessity
How is data shared	• Can be shared between police departments at their discretion • Geographic radius • Statewide • Nationwide
What does Flock Safety prohibit?	• Selling, publishing, or disclosing LPR data for commercial purposes • Disseminating data to unauthorized persons • Disclosing data without authorization unless required by law
What happens to data after primary use?	Less than 1% of data is used for machine learning purposes [11]

Figure 11. Summary of data management based on Flock Safety LPR Privacy Policy [8] and from the interview with GTPD Captain R. McNevin (group interview, Feb. 25, 2026)

Data Flow Diagram

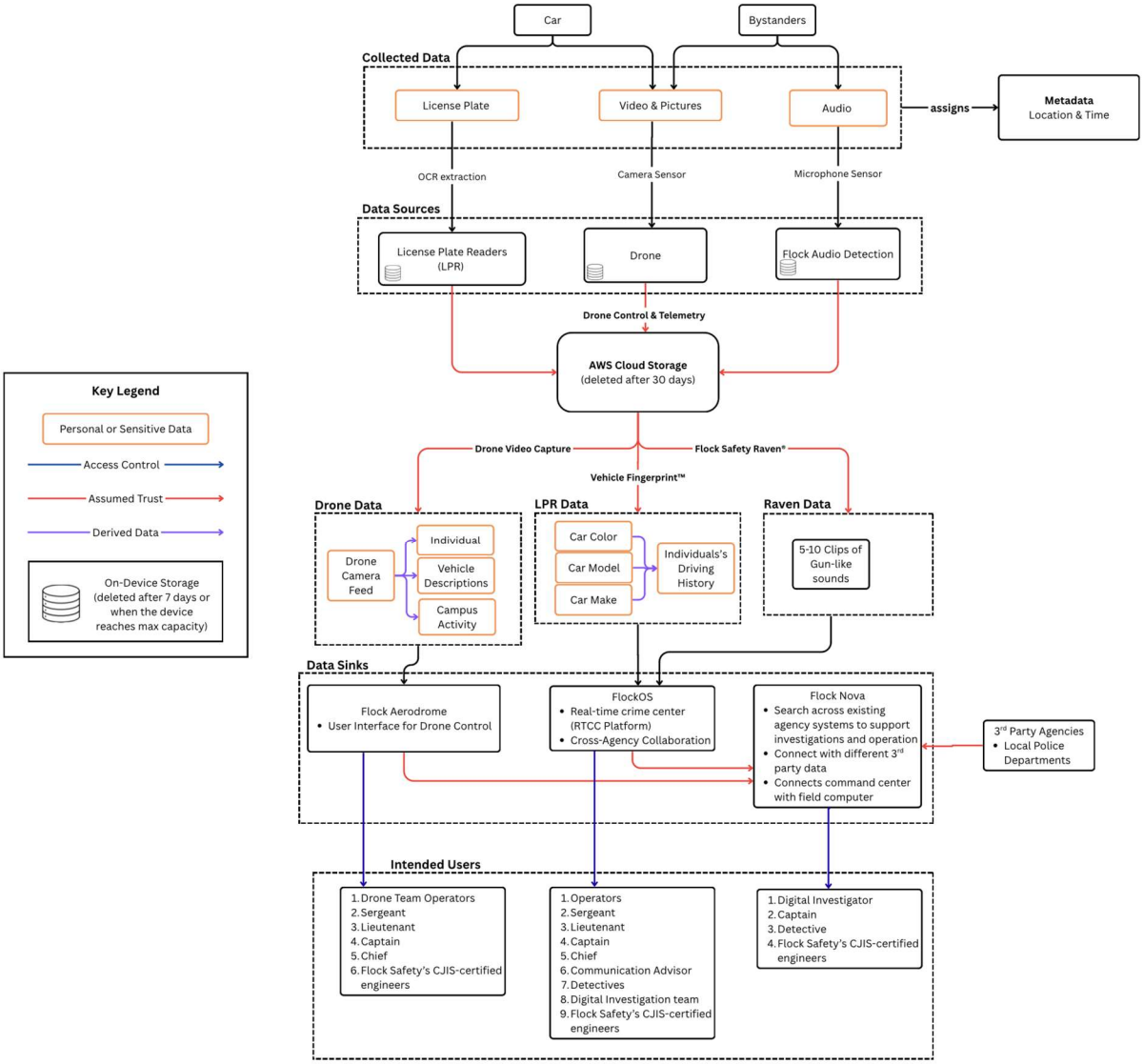


Figure 12. How Data Flows through Flock Safety Ecosystem

Data Sinks are defined as the final destinations of data, which we define to be the platforms that hold that data to be used for some purpose.
Intended Users are defined as people authorized to engage with the provided services for a specific purpose.

Technical Risks

Data Storage & Retention

Data Storage
Flock Safety stores their data using AWS cloud servers on an S3 bucket using AES256 encryption and transmitted with TLS 1.2, according to their privacy policy.

"Footage is collected on Flock's devices and transmitted from devices to Amazon Web Services using encryption in transit and stored within an Amazon's Simple Storage Service (S3) (organized according to customer lifecycle needs) for thirty (30) days, or as long as required by applicable law or regulation, and then permanently deleted. Direct access to this data is managed through AWS Identity Access Management policies and roles. Customer Data is encrypted throughout the entire lifecycle and transmitted using TLS 1.2 or better and stored securely using AES256 encryption. Customer Data collected on Flock's devices is stored locally until the device reaches storage capacity, or 7 days, whichever is sooner. While stored in the cloud, all data (both footage and metadata) is fully encrypted at rest using AES256" [11].

For more sensitive data such as in Criminal Investigations, they use AWS GovCloud with Key-Managed Services (KMS), meaning that Flock Safety has access to the keys as opposed to Amazon. AWS GovCloud has much higher requirements to meet regulatory and compliance needs for customers and utilizes cryptography standards authorized by the federal information processing standards. AWS GovCloud operates on a 'shared responsibility' philosophy [9][10]. As Flock Safety is a customer of AWS GovCloud, they are responsible for the Customer portion in Fig. 13. However, it should be noted certain individuals are authorized to access this information, as noted in their privacy policy [11].

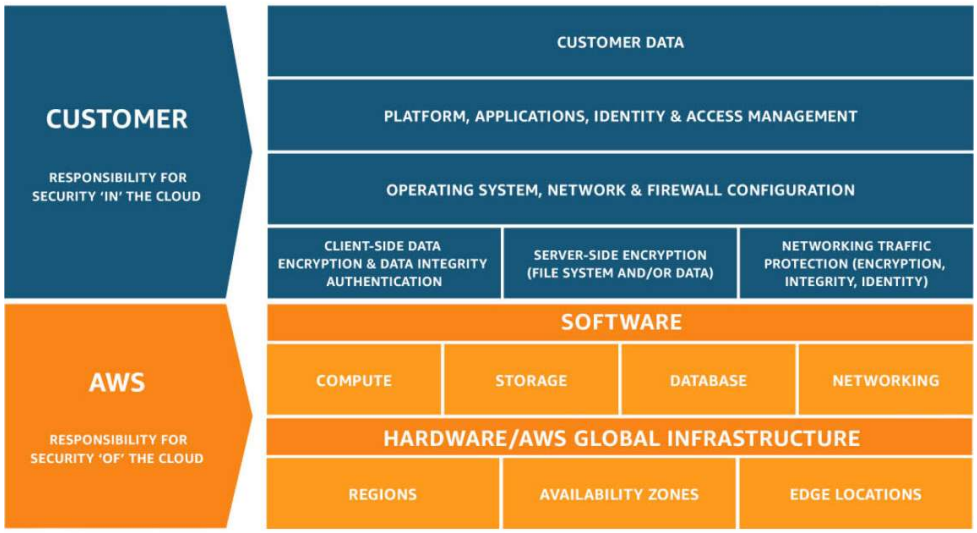


Figure 13. AWS GovCloud 'Shared Responsibility'

Data Retention

Their default data retention is 30 days on their cloud servers on AWS. The data is stored locally on devices for 7 days or until the device reaches capacity. Afterwhich, it is sent to their AWS servers on the cloud. Other specific regions adjust their own default data retention policies if this differs from what they want, so they respect existing LPR retention laws and regulations.

If they want a longer retention law, they have to get approval from a democratically elected official. Additionally, this can be extended if necessary for a criminal investigation as such required by a subpoena. This information is outlined in their evidence policy [12].

R1. Data Storage & Retention Risk Assessment

Component	Rating	Justification
Impact	Severe	A breach of centralized vehicle tracking and criminal investigation data could lead to systematic profiling or the exposure of highly sensitive PII.
Likelihood	Rarely	Strong, verifiable technical procedures (AES256 encryption, TLS 1.2, strict IAM policies, and AWS GovCloud KMS) are in place, requiring complex exploits to bypass.
Final Risk	Medium	

Artificial Intelligence (AI) and Machine Learning (ML) Algorithms

Flock's Vehicle Fingerprint®	Identifies vehicles by their make, model, color, and unique characteristics such as damages or accessories.
Flock FreeForm	Searches existing video and LPR systems with a natural language query.
Flock Safety Raven® Audio Detection	Provides real-time alerts using acoustic signatures and machine learning, provides location of gunshot with up to 90% detection accuracy and location accuracy up to 90 feet [5].

Figure 14. An overview of Flock Safety AI/ML algorithms with a brief description

Flock's Vehicle Fingerprint®

Flock's Vehicle Fingerprint system identifies the license plates of vehicles using OCR and Machine Learning algorithms, as shown in our Data Flow Diagram in Fig. 12. Additionally, this technology collects a vehicle's make, model, color, and the unique characteristics in order to assist with identifying the vehicle. Universities utilize this technology with their LPR cameras in order to support public security concerns. Georgia Tech, for instance, as of February 2026 has 15 LPR cameras according to our interview with Chief Connolly.

Their police department claims that this technology only captures license plate information, however the information by Flock Safety and in our Data Flow Diagram in Fig. 12 describes that it has the capability to collect much more information than just license plates such as car model or car make. Every university seems to have different policies on this. However, it does not track personal identifying information such as the name or address of the individual. In order to access this, you would need to go through state centers such as the Georgia Crime Investigation Center (GCIC) to identify a person with the license plate, which is outside the jurisdiction of Flock Safety.

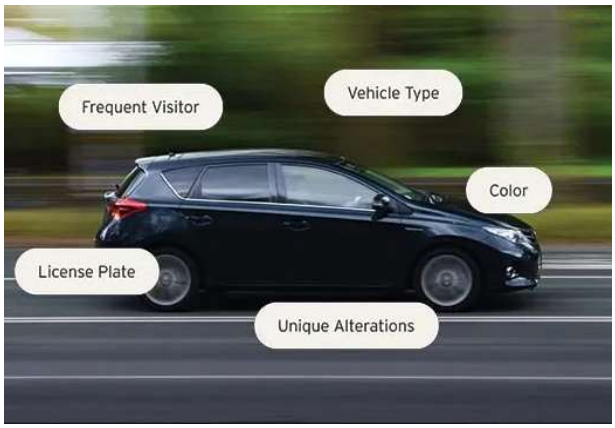


Figure 15. The data that Flock Safety LPRs can capture [13]

R2. Flock Vehicle Fingerprint Risk Assessment

Component	Rating	Justification
Impact	Severe	Tracking unique vehicle characteristics and movement history enables systematic profiling and "linkage attacks" to identify individuals, even without direct name or address data.
Likelihood	Almost Certain	The system defaults to maximum data collection (capturing make, model, color, and unique marks) as a core function of the technology.
Final Risk	High	

Flock FreeForm

Flock FreeForm is meant to make search easier through the collection of data from Flock Safety by using natural language queries in order to identify certain vehicles. However Flock Safety says specifically “No. FreeForm can search based on people characteristics and behaviours, but it is not facial or person recognition, and does not infringe on an individual’s right to privacy.” [14] But through looking at the capabilities, this neglects the privacy risks. In a feature of this product, you can identify people based on specific characteristics such as “person wearing a blue shirt and a cowboy hat”. This can uniquely identify people with certain characteristics, which impedes on their privacy.

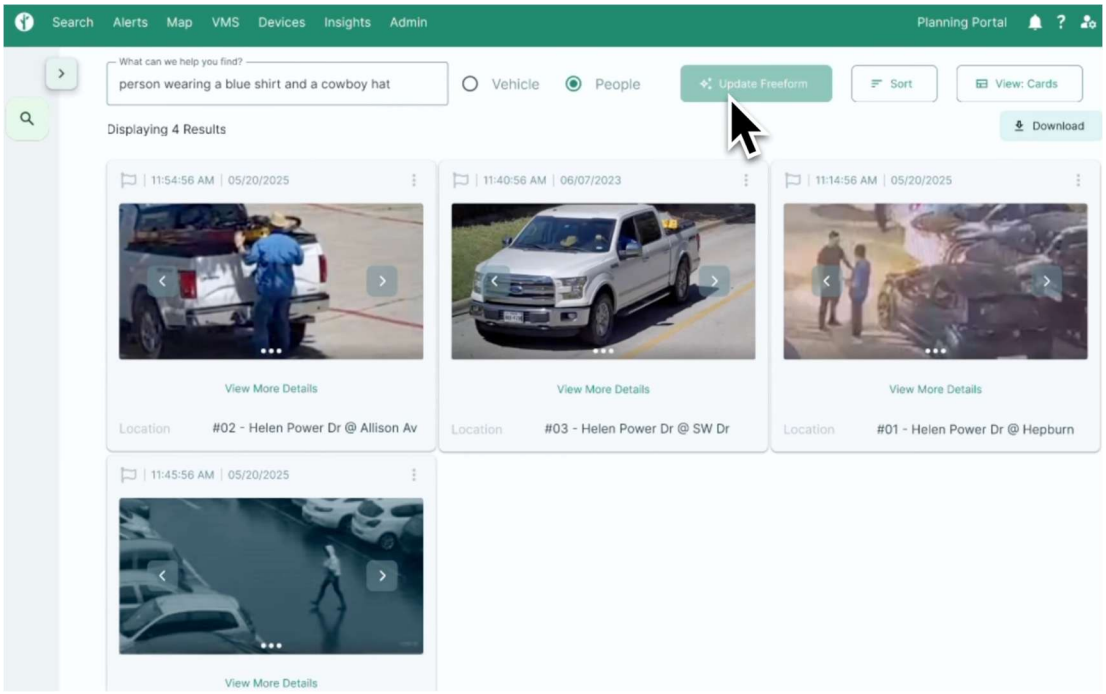


Figure 16. The Flock Freeform dashboard

Additionally, they acknowledge the need for the use of responsible AI so they use a content moderation philosophy based on “freedom of speech”, “religion”, and “assembly” but details on the specifics of their content moderation are not available [14].

Additionally, there exist certain search filters based on the legislation of specific regions and their safeguard access to that data [15]. This allows the ability of searching to be specific to the privacy legislation of that specific region’s jurisdiction. While a university like Georgia Tech doesn’t use this feature, it is unclear how easy it is to enable the entire ecosystem to use existing data.

R3. Flock Freeform Risk Assessment

Component	Rating	Justification
Impact	Severe	Allowing natural language searches for specific physical traits (e.g., clothing, behavior) bypasses general anonymity and enables systematic profiling of individuals.
Likelihood	Frequent	While mitigations like regional search filters exist, they rely on user configuration and can easily be bypassed or misconfigured, and their moderation policies are vague.
Final Risk	High	

Raven Audio Detection

Raven Audio Detection collects audio clips from the surrounding environment based on audio that sounds like gunshots and then alerts customers of any gun-like shot, directly sending them the audio clip. In a webinar introducing this technology, an employee explains that Raven captures 5-10 second audio clips of gunshot-like sounds, then sends them to the cloud for better processing, and then deletes them locally. If the hardware has more than 10 audio clips, it is no longer able to record any more data.

The hardware specifically has an onboard low-power machine learning model to detect and classify for gunshots, then uploads to a larger cloud to do additional processing. They claim it is not possible for someone to remotely tap in to listen to any conversations. However, there is a possibility of the product collecting gun-like shots along with conversation as described where 2 people’s conversations are heard over gun shots with individuals saying ‘hand me to the gun’. This is described as rare. A Figure of the Raven ML process is shown in Fig. 17, along with the data flows within Raven in our data flow diagram in Fig. 12.

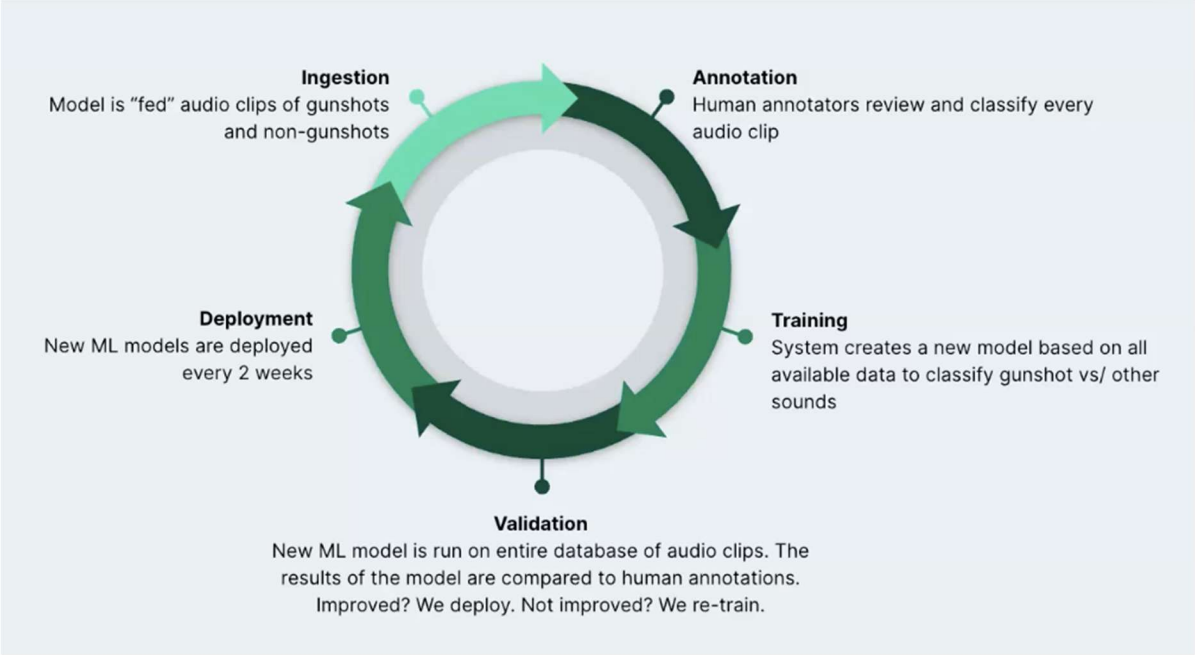


Figure 17. Raven Audio Detection ML Workflow [16]

R4. Raven Audio Detection Risk Assessment

Component	Rating	Justification
Impact	Moderate	While the data scope is limited to 5-10 second clips, captured audio could unintentionally contain private conversations occurring near a triggered event.
Likelihood	Rarely	The system relies on specific acoustic triggers and strict hardware limitations (e.g., maximum of 10 clips), making the incidental collection of private conversations highly unlikely.
Final Risk	Low	

Data Handling for ML Workflows

Data Ingestion Pipeline

Flock Safety takes their images from their cloud servers, extracts the metadata, and feeds that metadata into a third party, Clickhouse [17]. This metadata includes vehicle type, make, and license plate state from the images that Flock Safety captures using their products.

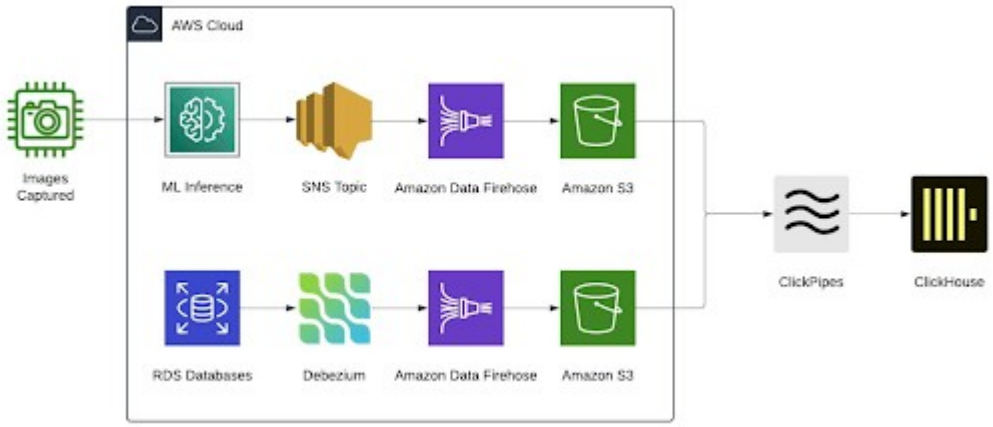


Figure 18. Clickhouse Analytics

This third party handles a lot of data. The article says that "At its peak, ClickPipes processes over 20 MB per second, allowing Flock Safety's traffic analytics platform to handle over 1 billion ML predictions per day with zero downtime" [17]. This leads to a high privacy risk as this metadata is out of the jurisdiction of Flock Safety.

R5. Data Ingestion Pipeline Risk Assessment

Component	Rating	Justification
Impact	Moderate	Metadata is shared with third-party analytics (Clickhouse), moving it outside of Flock's direct jurisdiction that can possibly identify individuals.
Likelihood	Almost Certain	This pipeline is a core technical requirement for the platform's traffic analytics to function.
Final Risk	High	

Training Data

Flock Safety's privacy policy notes the following:

"Training Data: We may use a small fraction (less than 1%) of images captured by Flock Services, which are stripped of all metadata and identifying information, for the limited purpose of improving our products and services through machine learning ("Training Data"). Training Data is used internally to help our systems better recognize vehicles, objects, and descriptions. For example, if a new Red Ferrari is released, we may use de-identified images of that vehicle to teach our system to accurately distinguish it from other vehicles, such as a Red Lamborghini. These images are never sold or shared with third parties. To protect your privacy, Training Data

is maintained separately and is never combined in a way that could render it personally identifiable. Use of Training Data is essential to ensure the accuracy and effectiveness of our services." [11]

It specifically explains the use of less than 1% of data for training their models. However, it is not clear how this data is randomly selected. If used for outliers to improve the model, such as a Red Ferrari, this implies there may be choice in how that data is selected. Additionally, Flock Safety collects 20B scans of license plates monthly according to their website. If they take multiple images in order to scan these license plates, along with other detection systems, then 1% of 20B is still 200 million, which is a massive amount of data. In the article with ClickHouse, they say that Flock Safety's traffic analytics handles over 1 billion ML predictions per day, which is only increasing with the amount of deployments they have.

Additionally, the information of their terms and conditions explain:

"Customer hereby grants to Flock a limited, non-exclusive, royalty-free, irrevocable, perpetual, worldwide license to (a) use and disclose Customer Data to provide the Flock Services; and (b) use Customer Data to support and improve Flock's products and services." [18]

This means that customer data can be used for training pipelines indefinitely, which is shown to be fed into these third parties for analytics. Ultimately, 1% of training data leads to a massive amount of metadata for vehicles, and they have the indefinite license to train, possibly using third parties, and there's no clear policy on how that training data gets chosen.

R6. Training Data Assessment Risk

Component	Rating	Justification
Impact	Severe	Involves a massive volume of data (potentially 200M+ scans) held under perpetual licenses.
Likelihood	Almost Certain	Terms of service grant Flock irrevocable rights to use customer data for training by default.
Final Risk	High	

Cybersecurity Risks & Company Response

Cybersecurity Vulnerabilities

Flock Safety creates embedded devices that store data, which creates opportunities for cybersecurity risks for their systems leading to serious privacy risks.

Experienced security researchers have looked into Flock Safety's ecosystem and have released detailed reports on the potential vulnerabilities identified within their products. We use CVEs from the National Vulnerability Database reported on Flock Safety in order to detail what privacy risks these technical risks create. We categorize these risks into three categories: Unauthorized Access to Personal Data, Intercepted or Altered Personal Data, Unauthorized Public Surveillance. We choose a CVE characteristic of this risk, but there may be more relevant to the category, so this is not meant to be an exhaustive technical list. More information can be found in the following resources of the cybersecurity research report [19].

Table 1. Unauthorized Access to Personal Data

Privacy Impact	On local devices, they are able to get data because of a lack of encrypted data. As this holds vehicle data, this can then have other algorithms that may identify people.
Technical Reason	These devices feature insecure hardware configurations, specifically unlocked bootloaders and unrestricted low-level diagnostic modes (EDL). This allows an attacker with physical access to bypass Android Verified Boot (AVB) and directly dump or modify the device's local memory and data partitions.
Affected Products	Bravo Edge AI Compute Device.
Primary CVE	**CVE-2025-59404** (Unlocked bootloader allowing bypass of Android Verified Boot and partition modification) [20] **CVE-2025-59402** (Unrestricted EDL mode allowing physical attackers to bypass security and dump device memory) [21]

Table 2. Intercepted or Altered Personal Data

Privacy Impact	May be able to get data as it transmits from system to system, and as this information is used for crime investigations, may lead to discrepancies. This impacts people's privacy as they may be accused of things they aren't responsible for due to being part of this system.
Technical Reason	Critical security credentials, including Auth0 API secrets and Java Keystore passwords, are stored as cleartext within the application's compiled code. An attacker can trivially decompile these applications to extract the secrets, enabling them to intercept authenticated traffic or spoof system identities to alter data.
Affected Products	Falcon and Sparrow LPRs, Bravo Edge AI Compute Device

Primary CVE	**CVE-2025-59406** (Cleartext Auth0 client secret hardcoded in application code, risking unauthorized API authentication) [22] **CVE-2025-59407** (Java Keystore bundled with a hardcoded password, exposing private cryptographic keys) [23]

Table 3. Unauthorized Public Surveillance

Privacy Impact	May be able to set up opportunities for someone to publicly surveil.
Technical Reason	Production firmware contains hardcoded development Wi-Fi credentials and unauthenticated administrative API endpoints (Port 8080). This allows unauthorized users in proximity to join the device's network and execute remote commands, granting them access to live camera feeds and system logs.
Affected Products	Falcon and Sparrow LPRs; Bravo Edge AI Compute Device (Collins app).
Primary CVE	**CVE-2025-59409** (Hardcoded development Wi-Fi credentials in production firmware, allowing unauthorized network access) [24] **CVE-2025-47823** (Hardcoded system password in LPR devices allowing unauthorized access to system functions) [25] **CVE-2025-59403** (Unauthenticated administrative API endpoints on port 8080 exposing logs and camera control) [26]

R7. Cybersecurity Risk Assessment

Component	Rating	Justification
Impact	Severe	These vulnerabilities enable the access towards sensitive data of individuals and then individuals ability to use their own algorithms on the data, rather than just Flock's de-identified algorithms.
Likelihood	Frequent	It is very difficult to execute these but it is not clear if these vulnerabilities are being fully addressed.
Final Risk	High	

Flock Safety Response to Security Concerns

Flock Safety has responded to the compilation of these security vulnerabilities explaining that they've done a thorough analysis explaining "Overall, none of the vulnerabilities detailed in the report have an impact on our customers' ability to carry out their public safety objectives.". They explain this is because the knowledge required to exploit these vulnerabilities require physical access so the risk is low. However, this perspective neglects the privacy objectives [27].

Additionally, they maintain that "Flock has never been hacked". They do not believe these theoretical attacks to constitute a 'hack' as there has not been reported incidents that act on these vulnerabilities [28]. However, the amount of vulnerabilities to unauthorized sensitive data reveal a lot of gaps in their cybersecurity practices, which then lead to privacy risks. They are weak in cybersecurity but they've recently hired Chief Information Security Officer Chris Castaldo who has extensive cybersecurity experience [29]. Additionally, they've signed an engagement with Bishop Fox to boost their cybersecurity [30].

Ultimately, cybersecurity risks become privacy risks if they are not addressed. A theoretical risk is the first step towards real exploitation.

R8. Flock Safety Response to Security Concerns Risk Assessment

Component	Rating	Justification
Impact	Severe	Flock Safety's response to these cybersecurity risks are alarming, however, they are taking the steps towards filling this gap.
Likelihood	Frequent	While their philosophy is alarming, they are taking steps towards cybersecurity.
Final Risk	High	

Integrations & Third Parties

Flock APIs & Integrations

Flock Safety owns a developer portal for customers to build integrations with other services using their APIs. This allows customers to use Flock Safety data with services in CAD or VMS, as shown in Fig. 3 [31]. These integrations are easily accessible and span multiple categories. However, since this data goes towards other partners, there is a privacy risk because the handling of this sensitive data is now under the jurisdiction of another company, even though Flock Safety originally captured this data. One of their partners, Genetec, has specific documentation outlining the integration with Flock Safety [32].

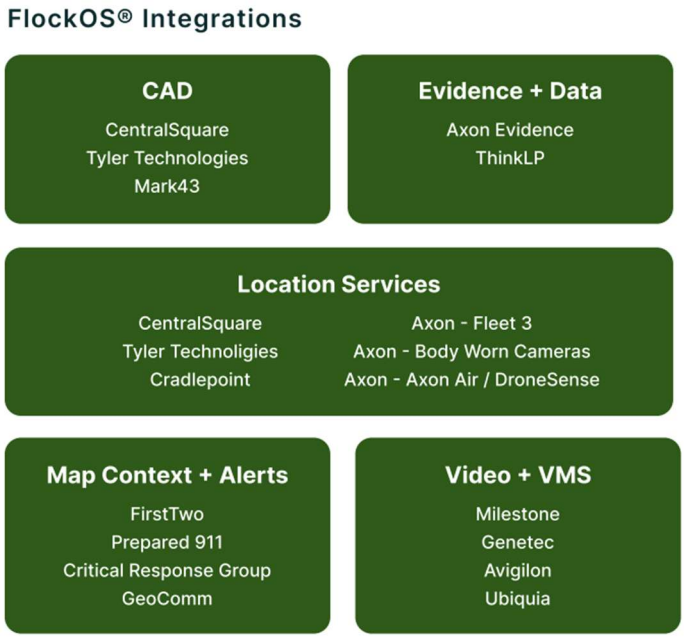


Figure 19. Flock integrations

Their API and Integrations Terms outline their standards to mitigate these risks. They specifically focus on Auditability where people must request reasons for queries and Data Stewardship to ensure that this API access is for real-time workflows to ensure responsible data handling and prevent database replication or data scraping [33]. This limits the possible privacy risks, but it is unclear how exactly they ensure these principles on a technical level.

R9. Flock Safety Third Party Integrations Risk Assessment

Component	Rating	Justification
Impact	Severe	The data originally collected by data is now open to other third parties which could potentially store them under different regulations than originally intended by the community adopting Flock.
Likelihood	Frequent	APIs and integrations aren't used by everyone, and they have strong terms in how these APIs are used.
Final Risk	High	

Drone First Responder

Flock Alpha

Drones are used in the Flock Safety ecosystem to quickly respond to incidents. They have HD cameras, thermal cameras, and color night vision. Their Drone First Responder uses Flock Alpha as their hardware product. Georgia Tech specifically uses the DJI Drone M4TD RC PLUS 2 as reported by the Drone Team of the GTPD (group interview, Feb. 26, 2026). However, the firmware is not DJI and only Flock Safety's proprietary firmware.

They maintain waivers with the FAA to fly at higher altitudes than previous jurisdiction-wide approvals. Additionally, they specifically partner with O.W.L. Radar systems to develop their capability through 3D radar systems. This allows the Flock Aerodome system to support the use of remote piloting and uses radar-based detect-and-avoid (DAA) technology to avoid crashing into other aircrafts autonomously. This along with ADS-B, Remote ID, and RF Detection, enable it to have a good understanding of the airspace [34]. No exploitations of these systems have been found. This collects data of the surroundings as a result of the radar, from a much higher altitude, which allows them to see more.

Drone Software & Data Collection

Their Drone software is Flock Aerodome. It utilizes a keyboard and mouse to control the drone, enabling beyond visual line of sight (BVLOS) operations (Drone Team Member, group interview, Feb. 26, 2026).

The drones provide GPS coordinates to the CAD system, connect to other existing systems, share real-time alerts, and can be autonomously deployed. Drones can start live streaming video and thermal imaging. This leads to a fast response. An example is an LPR camera picking up the plate of a stolen vehicle, and flock safety cameras can send GPS coordinates and an image of the vehicle to the drone systems [35]. This then stores GPS coordinates based on certain vehicles, creating more potential privacy risks.

These drones are controlled by a team, have no facial recognition, and have specific policy/SOP guidance for transparent, ethical use. They fly only for specific security purposes. [36]. For instance, Georgia Tech requires you to fill out a form with the reason for the use of the Drone (Drone Team Member, group interview, Feb. 26, 2026).

However, it impacts privacy because it impedes on students and they may believe there is mass surveillance as a result.

R10. Drone Alpha Risk Assessment

Component	Rating	Justification
Impact	Severe	The drones are only meant to respond to incidents but they do store information and retain that in a way that can see a lot more than just LPRs.
Likelihood	Frequent	The drones are used for incidents but they can fly in any way possible so they can collect any information.
Final Risk	High	

Societal & Ethical Risks

Hardware Expansion

Flock LPR Expansion

From the interview with Captain McNevin (group interview, Feb. 25, 2026) and Chief Connolly (class interview, Feb. 23, 2026) Georgia Tech is possibly planning on expanding the usage of Flock devices, such as LPR, audio detection, and drones due to the growing amount of construction work around campus. Yet, the expansion is very limited since Flock devices are only beneficial in areas where mobility and security are key; therefore, Georgia Tech limits the installation and usage of Flock devices (Chief Connolly, class interview, Feb. 23, 2026). However, if there were an expansion of the use cases for these hardware devices used by Georgia Tech Police Department (GTPD): LPR, audio detection, and drones, there would be varying risks.

For the expansion Flock LPR device, this would be a low risk. Flock states that their LPR "observe vehicles, not people. They read license plates and vehicle characteristics that are meant to be publicly visible. From these images, one cannot know who drove, whether drivers changed, who rode as passengers, or what anyone did before, during, or after an ALPR's roadway observation [37]." However, images can possibly include people or identifiable information on an individual while taking a picture of the license plate. While we were not able to locate information on whether Flock LPR devices may include pictures of people, we found an example of the Flock LPR device pictures below.

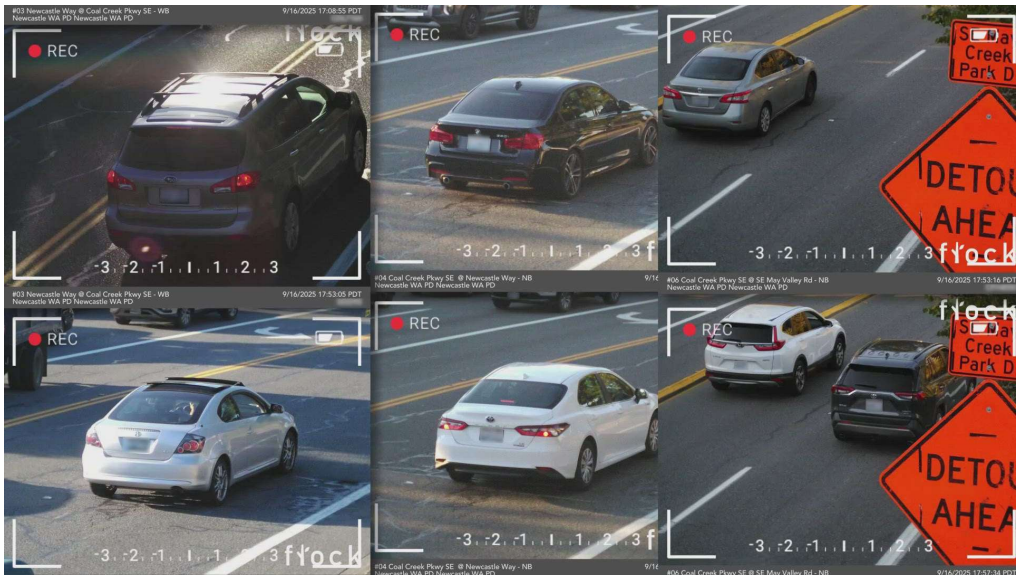


Figure 20. KING 5 News Flock LPR Images [38].

This shows that although Flock LPR readers do not specifically identify people, its field of view can possibly capture people outside of the license plate, risking personal identifiable information to be shown if the LPR readers are installed without taking this into consideration.

R11. Flock LPR Expansion Risk Assessment

Component	Rating	Justification
Impact	Moderate	It is possible for Flock LPR cameras to possibly capture people outside of the license plate, which is dependent on how Flock focuses the pictures to process only the vehicle and how the Flock camera is installed and set up during the expansion.
Likelihood	Rarely	The scope of the LPRs focus mainly on vehicles and license plates, so it would be hard to capture people outside of that.
Final Risk	Low Risk	

Flock Audio Detection Expansion

For the audio detection device expansion, this would be a high risk. The reason why the expansion is high risk is not because of the increased number of audio devices, but the range that the audio captures. According to Flock, they can capture a range of noises from gunshots to "disruptive community activity [39]." Currently, GTPD only uses the audio detection for gunshots; however, if the expansion of the sound types are used, this would be a high risk. This is

because the AI models are not exposed to every type of noise possible and can hallucinate the type of sound, resulting in multiple false positive recordings, which was shown with a different security company used at the University of Michigan [40]. Therefore, this is a high risk expansion in terms of what sounds GTPD would want to capture.

R12. Raven Detection Expansion Risk Assessment

Component	Rating	Justification
Impact	Severe	Expansion of this technology could increase the amount of audio information captured on individuals at Georgia Tech which raises the risk of PII that GTPD could obtain.
Likelihood	Frequent	Expansion is more possible due to the Raven Detection Audio devices being portable, easily able to purchase, and simple to install
Final Risk	High	

Drone Expansion

Currently, in the drone system used by GTPD, they limit the drone's usage only to respond to gunshots (Drone Team Member, group interview, Feb. 26, 2026). In addition, the GTPD drone does not have the capabilities of autonomously deploying to events and patrolling scheduled paths in the interview with the Drone Team Member (group interview, Feb. 26, 2026). However, with the expansion of a new drone developed by Flock called Flock DFR (Drone First Responder), this would unlock new capabilities, such as patrolling autonomous paths and deploying automatically. The risk with this expansion is the pilots who are piloting the drone. However, there is an extensive training process with Flock and GTPD with a previous background knowledge in drones recommended, flight summaries that are required to be filled after every flight, and flight recordings, which record the whole flight automatically (Drone Team Member, group interview, Feb. 26, 2026). Another risk is the type of information that the drone transmits, which are pictures and videos that may include personal identifiable information about people. Therefore, we have determined that the drone is a low privacy risk due to the accountability measures in place and information transmitted.

R13. Drone Expansion Risk Assessment

Component	Rating	Justification
Impact	Moderate	This expansion still will have multiple accountable checks in place with GTPD and Flock system, but still have personal identifiable information that can be transmitted.

Likelihood	Rarely	Expansion is unlikely because of the many audits before actually being implemented. In addition, the restrictions on airspace around Georgia Tech campus.
Final Risk	Low	

Software Expansion

Flock Nova Expansion

Another growing risk of expansion is Georgia Tech's usage of the Flock Nova Platform. Captain McNevin stated that certain GTPD members have access to the Flock NOVA software platform: digital investigators, detectives, and Captain McNevin according to our interview with Captain McNevin (Captain McNevin, group interview, Feb. 25, 2026).

At this time, there is no current plan on increasing the usage of the Flock Nova Platform; however, this PIA will address the risks of its expansion if it were to take in place. Flock Nova is an AI powered, cloud based software platform that unifies the data received by multiple sources in a police department. Flock would describe this as:

" Flock Nova, an all-in-one data intelligence platform built to accelerate investigations. With Nova, law enforcement agencies can instantly access all of their investigative data in a single platform—combining CAD, RMS, video footage, LPR data, and even open-source intelligence in one unified experience [6]."

This saves time and budget costs for any police agencies. Yet, this brings its own risks. Research has shown that all-in-one platforms are more vulnerable for data leaks [41]. Furthermore, over 45% of data breaches occur in the cloud with public cloud security incidents averaging $5.17 million per breach, resulting from feature misconfigurations and account compromises [42]. With all of an agencies' data funneled into one platform would raise enormous risk of personal identifiable information of students, faculties, and employers of Georgia Tech leaked.

Furthermore, in our Data Flow Diagram in Fig. 12, we show that currently at GTPD, multiple data sources exist; however, each data source corresponds to different softwares with different purposes. However, the increased usage of Flock Nova will congregate all the data toward one platform, making this easier for data to be leaked or misused.

Not only that, Flock Nova allows for integrations with third party data. Currently, most data breaches 59% of data breaches involve third party suppliers [43]. Flock Nova's ability to integrate with third party data allows for accessibility, but it will increase risks of getting cyberattacks from unsecure third party vendors. Therefore, this expansion is a medium privacy risk.

R14. Software Expansion Risk Assessment

Component	Rating	Justification
Impact	Severe	This expansion would bring new vulnerabilities enabling the access of many personal identifiable information of individuals in Georgia Tech in one centralized environment.
Likelihood	Rarely	Georgia Tech Police Department does multiple auditing based on the stringent policies at Georgia Tech, limiting this expansion.
Final Risk	Medium	

Deployment & Communities

According to Chief Connolly's interview, the GTPD is actively limiting the amount of Flock devices and systems that they are using (class interview, Feb. 23, 2026). Furthermore, from Chief Connolly's interview, he stated that they have 15 Flock LPR and a small amount of audio detection strictly for gun shots (class interview, Feb. 23, 2026).

The number of Flock LPR was supported by our research when we used DeFlock Map in order to figure out how many Flock LPR are located [5]. Due to its locations, which are mostly located on the outskirts of the Georgia Tech campus, and its capabilities, we have concluded that the addition of Flock LPR is not disproportionately affecting certain communities.

However, with the limited data on the number of audio detection and approximately where they are used on campus, we can not certainly conclude that this will disproportionately affect certain communities. In Chief Connolly's interview, he did state that these audio devices only provide sounds that mimic a gunshot and do not send conversations to the operators (class interview, Feb. 23, 2026). It is important to note that these audio devices can pick up other sounds as detailed in the risk expansion section [39]. In our Data Flow Diagram Fig. 12, we have shown that there is a possibility that the Raven Data can go from the Flock OS and transmit to the Flock Nova system, making it a privacy risk. Therefore, if this were to be included into the usage of the GTPD, it will disproportionately affect certain communities at Georgia Tech, specifically the edges of Georgia Tech where there is more disruptive community activity, which can pick up more bystander sounds that can be identifiable.

Yet this time, with the devices using onboarding processings to identify sounds and Chief Connolly stating that these audio devices are programmed to only detect gunshot violence, we

have determined that the audio detection system will not disproportionately affect certain communities at Georgia Tech.

Risks of Surveillance Normalizations & Power Imbalances

There are risks of surveillance normalization due to external pressure and events that have occurred. In the recent Brown University shooting, this prompted multiple schools to increase campus security, including more patrols and monitoring, such as at Harvard University [44]. Prior research shows that with more anxiety and fear around these events, risks of surveillance normalizations increase as more security is put in place on campuses to prevent another of these events happening [45], [46]. A similar example was with the Patriot Act passed by Congress after the September 11 attacks, which expanded government surveillance [47]. While these protections are put in place to help people, the increased usage can normalize surveillance and create a rising risk of overreach or abuse of power.

There exists a power imbalance between system operators and affected individuals because the affected individuals may not know that their data have been used or stored within the cloud by Flock. Therefore, transparency regarding data practices is important. Providing documentation such as our Data Flow Diagram in Fig. 12 which details how possible personal identifiable information can be transmitted will support accountability and demonstrate the safeguards and protective measures are in place.

Dark Patterns

Security-Centric Framing

Flock Safety's brand and marketing rely on a false dichotomy of 'Safety vs. Privacy' which builds a narrative that anyone not fully supporting Flock's efforts is 'Pro-Crime' or 'Anti-Safety'. This may stifle real critiques and does not lead to meaningful discourse surrounding potential privacy or ethical concerns. In an email to the Staunton police department [48], the CEO of Flock Safety, Garret Langley asserted activist action against Flock Safety and their customers are solely from those who want to weaken public safety and normalize lawlessness. This raises major concerns about if and how Flock dedicates time to resolving ethical issues coming from the public. From a privacy perspective, this framing reduces the perceived legitimacy of privacy concerns and can create reputational pressure against dissent. Without adequate public input and the ability to listen to those who have worries about how systems may impact their lives, Flock runs the risk of overlooking valid privacy concerns as they are lumped in with other critiques as "Anti-Safety".

This is uniquely prevalent in a university setting where campus deployment of Flock devices is presented as a way to keep students safe. This then enables university administrators to turn down critics of Flock's systems by simply referring to student safety, regardless of the validity of the

critique. This, in combination with the power dynamic of students and campus officials may cause students to be more wary of asking questions in fear of being labeled as “Anti-safety”.

R15. Flock Safety Security-Centric Framing Risk Assessment

Component	Rating	Justification
Impact	Moderate	The framing around Flock’s systems can make it harder for healthy dialogue surrounding possible privacy risks. This privacy harm is indirect but structurally significant.
Likelihood	Frequent	There are several instances of Flock framing critics and dissenters with dichotomatic language.
Final Risk	Medium	

Illusionary Control

Flock Safety has a feature enabling residents of a neighborhood or street to add their vehicle to a “SafeList” [49]. This feature is touted as a way to maintain resident privacy and give peace of mind to those you opt in to using it. The feature allows Flock system keepers to send a form to residents wherein they may put their license plate numbers and street addresses so that they are either listed as resident not included in search results [50]. While this feature does allow for more individual choice in how their information is used in Flock’s systems, its implementation may lead to residents believing their information is being completely not tracked after completing the form. However, the images of license plates are still stored for the retention period, as “SafeList” only prevents you from appearing in general search results. Similarly, since system administrators are required to send out the “SafeList” form to their residents, it operates on an opt-out default which puts the pressure on the individual to secure their information. Also, for the communities who are unaware or unwilling to send out the form, this feature is unavailable and its privacy benefits are lost.

On a similar note, the fact that residents are required to opt-out rather than opt-in can make it harder for individuals to understand how and by whom their data is being collected. This nudges the system towards collecting and sharing more data than is necessary. The Electronic Freedom Foundation argues that to protect users’ rights to privacy, data collectors must be set up to allow users to opt-in rather than opt-out so that they are able to understand exactly what kind of data is collected from them [51]. Flock’s array of systems does not follow this model even with their “SafeList” since a resident may not even know their license plates are being collected if their community leaders don’t give them adequate notice and therefore, they also cannot choose to opt-out of searches using “SafeList”. This obfuscation of individual agency, in combination with other dark patterns commonly used by Flock means individuals are not meaningfully informed about how surveillance occurs.

R16. Flock Safety Illusionary Control Risk Assessment

Component	Rating	Justification
Impact	Moderate	Residents may be misled about how their data is being treated thereby distorting user understanding and creating false assurances of privacy.
Likelihood	Frequent	There is a good chance some residents of a given neighborhood will misunderstand how the SafeList works but mitigations can be easily implemented
Final Risk	Medium	

Camera Design and Installation

The design and physical appearance of Flock's ALPR cameras are another subject of concern. The cameras are designed to blend into their surroundings and look like any other traffic camera so as to not compromise neighborhood aesthetics [52]. This misleads individuals and causes them to be unaware their license plates are being recorded. Because of their discreet design, citizens may easily overlook them or believe the cameras are simply used to enforce traffic violations. In addition to this, individuals are rarely informed that their community is using Flock cameras. This minimization of notice makes it harder for the public to be aware of how their law enforcement agencies are using Flock cameras. In the same way, the ease of setting up these cameras encourages agencies to adopt and install them quickly, which forces the public to consent to high-level surveillance without their knowledge and approval. The EFF argues how ease of access for these surveillance tools come with a hidden cost to the public and their privacy rights [53]. Even though the systems are easy to set up for the local law enforcement agencies, they are trading away their authority over how they are used, which may become a risk in the future.

This ease of installation creates another caveat for customers. If a customer decides to cancel their Flock partnership, they have no control over whether the cameras actually get turned off [54]. In Evanston, IL, the city terminated their contract with Flock and had their cameras removed only for them to be reinstalled days later in similar locations [55]. The ease of installation but difficulty of removal of Flock cameras can make surveillance seem even more inescapable as residents don't know if efforts to remove cameras will actually lead to changes.



Figure 21. A vendor reinstalling Flock cameras in the city of Evanston, IL [55]

Universities maintain their own specific issues arising from the physical design of these cameras. Due to the academic nature of university campuses, they often bring in large amounts of visitors and commuters who will likely drive into campus with their own personal vehicles. Because there is no requirement for posting notice, and the cameras are designed to be purposefully overlooked, visitors and commuters may have no idea their license plates are being tracked by university law enforcement, they have no way of knowing how their information is being collected and stored unless they have previous information about university policy. This further extends the idea of minimizing information regarding Flock's systems and can increase feelings of unavoidability and inevitability when it comes to surveillance.

R17. Camera Design Risk Assessment

Component	Rating	Justification
Impact	Severe	Individuals who have no knowledge of Flock's systems or its implementation in a certain area will have no idea they are being surveilled.
Likelihood	Almost Certain	There is no requirement for posting notice of these cameras and so there's not really transparency around them.
Final Risk	High	

High-Friction Sharing Architecture

Flock has many customizable options for sharing data among agencies such as 1:1 sharing, Statewide Sharing, and Nationwide Sharing [57]. This allows for agencies to have more control over how their community's data is shared to other law enforcement entities. Our Data Flow Diagram (Fig. 12) shows how data can be shared to third party agencies. For example, GTPD shares only on a case by case basis, as confirmed during an interview with Chief Connoly (class interview, Feb. 23, 2026), while UAPD shares information with Arizona law enforcement [58]. While this does allow for more flexibility amongst users of Flock's systems, its implementation could cause some privacy risks, especially when misconfigured by the customer. In order to enable national and statewide lookup, all that is required is a checkbox which may accidentally be turned on. For example, the city of Mountain View, CA, found their Flock systems had nationwide and statewide lookup enabled for three months before they realized, in which time 250 agencies ran over 600,000 searches on data collected by Mountain View's cameras [59]. While the Mountain View incident arose because of a failure to notice they had their sharing enabled, the ease of activation for these sharing programs can lead to increased sharing despite what the customer believes they have enabled. This can lead to increased data collection, especially from those institutions which may have stricter policies regarding how their communities' data is shared.

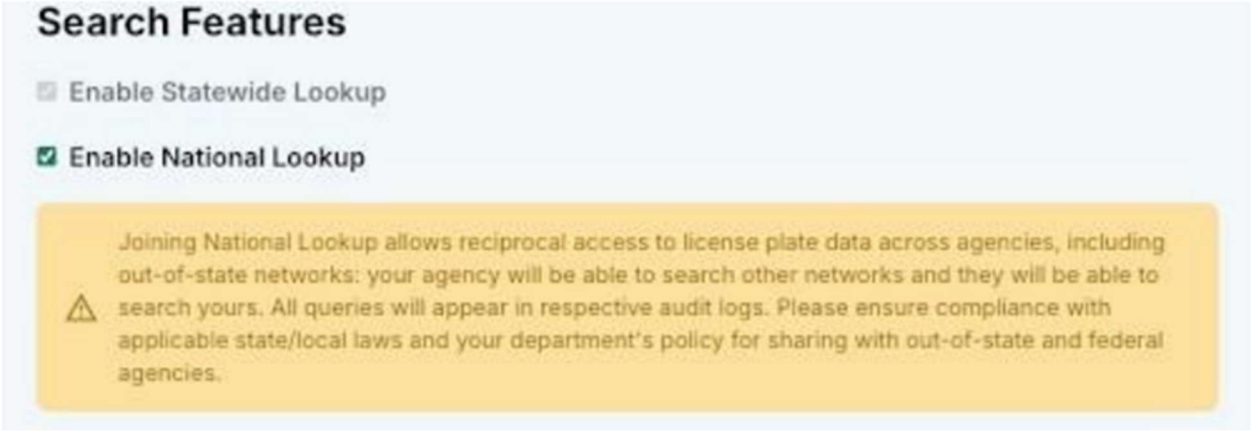


Figure 22. The checkbox to turn on National and Statewide Lookup [56]

R18. Sharing Data Risk Assessment

Component	Rating	Justification
Impact	Severe	Agencies who believe they are only sharing a certain amount of data may be unknowingly sharing a larger range through a configuration error.
Likelihood	Rarely	Changing the data sharing configurations is not something that should come about in general usage of the systems.

Final Risk	Medium

Chilling Effect on Activism

Universities campuses are often popular spots for activism and protests. Flock's systems may hinder this freedom of speech either by directly allowing law enforcement to collect data on who arrived at an event by car or by indirectly giving potential protestors more apprehension about showing up out of fear of being tracked. The EFF was able to identify 19 agencies across the country which ran searches related to the No Kings Protest in June and October 2025.

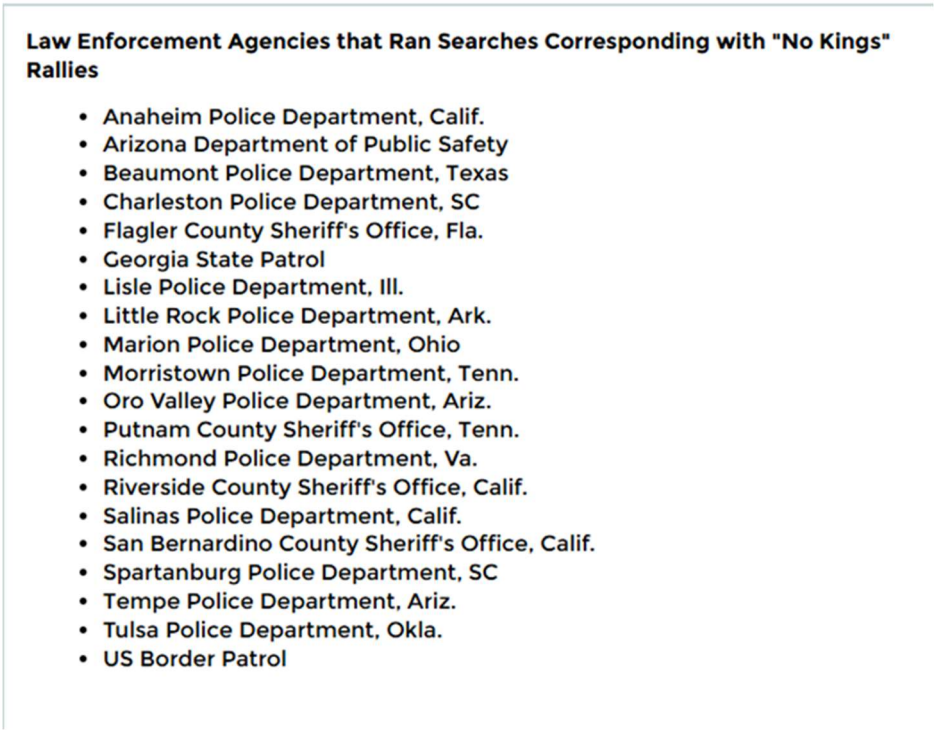


Figure 23. The EFF's compiled list of agencies that ran searches related to No Kings Protests [60]

Many of these searches used a search reason of simply "protest" or "no kings". While crime does occur at protests, and most of these searches were likely tied to actual cases, the lack of reasons makes it difficult to tell. The EFF also highlights how Flock systems were used to target animal rights activists from Direct Action Everywhere [60] showing how Flock's systems can be used to suppress activism. The idea that everyone who attends a protest could be surveilled creates a kind of chilling effect on activism which risks stifling constitutionally protected speech and compromises the privacy of activists.

This chilling effect is further intensified by the fact that many university law enforcement agencies are not elected by those they serve, but rather appointed by administration, thereby giving students and faculty a sense of inevitable surveillance as they have little say in how their university chooses to adopt safety measures. Although many universities require protests to be approved by administrators beforehand, they generally do not require a list of all the attendees, which is something that may be found out through Flock's systems.

R19. Flock Safety on Activism Impact Risk Assessment

Component	Rating	Justification
Impact	Minor	There isn't as much of an expectation of privacy at a public protest, however the chilling effect created by Flock's systems may have other ethical and legal concerns.
Likelihood	Rarely	There are multiple instances of protestors being tracked using Flock's systems.
Final Risk	Low	

Risk Mitigation Strategies

Overview

Flock has many security certifications and privacy safeguards relevant to information security and privacy governance [61]. These frameworks address many aspects of Flock's data ecosystem but do not replace a university's own legal and policy obligations, they provide a baseline indicating how Flock is subject to controls over their handling of sensitive information and demonstrate how independent auditing is done.

CJIS

Flock is CJIS certified through AWS GovCloud [62]. This ensures all criminal justice information is stored and encrypted securely using AWS KMS keys. This ensures even if their databases were compromised, the data would remain unreadable. Access to their databases is controlled and limited only to employees who have completed security training and a fingerprint based background check. Compliance with CJIS also certifies data is not shared to any external sources. This safeguards the data collected by Flock and makes sure only those who are authorized are able to access the data collected by Flock's systems.

FedRAMP 20x

Flock Safety meets the requirements for the Federal Risk and Authorization Management Program 20x pilot. This is a federal authorization which provides a standard approach to security assessments for cloud computing services [63] . This indicates Flock's abilities to handle confidential and encrypted information in a cloud computing setting. FedRAMP 20x certification shows how Flock is able to use cloud computing to securely work with law enforcement data.

NDAA compliance

Compliance with the National Defense Authorization Act underlines how security hardware is sourced from specific manufacturers. This act is imperative for securing federal funding for federally funded projects in the US. While Flock is a private company, it requires compliance with this act in order to conduct business with federally funded institutions, as many universities are. Compliance also ensures heightened security when it comes to national security threats and can help prevent leaking private data to external entities.

ISO/IEC

There are five ISO/IEC standards Flock Safety complies with including: **ISO/IEC 27001, 27017, 27018, 27701, 42001:2023**. These standards corroborate Flock is able to operate a formally audited security management system, define cloud specific security practices, protect PII in the cloud, address governance of personal data, and follow a framework for AI systems [64]. By following these information security practices, we can assert Flock is able to follow frameworks which support confidentiality, integrity, and availability of data as well as has structured processes regarding privacy obligations. They also show us Flock can store and manage large amounts of personally identifiable information without risk of compromising data. In general, compliance with these standards ensures Flock is able to adhere to internationally recognized information and cloud computing security standards.

SOC 2 Type 2

This certification ensures an independent auditor has tested the design and implementation of Flock's systems over a period of 6-12 months [65]. This audit can help to provide assurance that data is protected over a longer period of time as well as help to identify potential weaknesses in how the organization makes decisions regarding risk management. Overall, this certification helps customers understand Flock is following their stated data retention periods and can help to reduce the impact of a potential data breach.

Data Retention Limits

Flock has a policy of retaining data for no longer than 30-days unless flagged in a current investigation [8]. While this upper limit can be extended on a case-by-case basis, it rarely is as it requires approval from an elected official or governing body. This also limits the ability of Flock to be used for long term tracking and serves as a form of automatic data minimization.

No Sale of Personal Data

Flock states it does not sell collected ALPR data to data brokers nor facilitate the dissemination of ALPR data to unauthorized individuals [8]. This also prevents Flock from giving over data collected by the customer to other customers without their approval. This choice enables Flock to mitigate the risk of function creep into the commercial surveillance market.

Resident SafeList

This feature allows for customers, mainly HOAs and neighborhoods to send out a form to their residents to add their vehicles to a Safe List [49]. License plates on the safe list can be registered either as 'Resident' or configured not to show up at all in a search. This allows for residents to, in a way, opt-out of Flock's systems and provide peace of mind that their vehicles won't show up in a criminal investigation.

Recommendations

Data Protection Recommendations

In our risk assessments, we have determined that **R1**, **R2**, **R3**, **R5**, **R6**, **R8**, **R9**, **R10**, **R12**, **R13**, **R14**, **R18**, and **R19** have risks associated with data. **R1**, **R2**, **R10**, and **R14** are vulnerable with PII data being stored and misused on a server or a centralized platform. **R3**, **R5**, **R6**, **R9**, **R10**, **R18**, and **R12** are vulnerable with PII data being shared by other platforms, especially data used in Artificial Intelligence training and inference. Therefore, we have strongly recommended the following data practices to be implemented in Flock's deployment at Universities such as Georgia Tech.

- First data practice is to incorporate data masking and hashing, which transforms sensitive information into realistic but non-identifiable formats [66]. This masking must be applied specifically when the data enters the AWS Cloud Storage before the LPR data flows into downstream Data Sinks like Flock Nova or Flock OS as detailed in our Data Flow

Diagram in Fig. 12, which ensures that any PII data transmitted through data sources are not traceable when used in AI training or data collection.

- Furthermore, the NIST framework provides a guide on Protecting the Confidentiality of PII, which should be used as a guideline in order to protect information, and documentation for Quantum resisting cryptographic algorithms that will protect data from being stolen and deciphered later [67]. Finally, using new research practices, such as Adaptive Encryption will further increase the protection of the data [68].
- Also, we recommend a strict security clearance and access control of data. This recommendation is more centered toward the GTPD and the **Intended Users** outlined in our Data Flow Diagram in Fig. 12. Research has shown that hierarchical roles and zero trust architectures prevent cyberattacks from occurring [69]. In the context of our Data Flow Diagram, this means enforcing strict, role-based access boundaries; for example, ensuring that only a Communication Advisor or Senior members of GTPD (e.g., Lieutenant, Captain, Chief) only has access to high-level Flock OS alerts, while restricting sensitive Data Sinks like Flock Nova or raw Drone Camera Feeds exclusively to detectives, the digital investigation team, and senior leadership. This would help prevent unauthorized access of user data and reduce the insider threats occurring within an organization. By implementing such architectures directly at the access control points of these distinct data sinks, GTPD can more securely protect user data.

Training Recommendations

While assessing risks **R11**, **R7**, and **R15**, we found that these risks can be mitigated with more in depth or comprehensive training programs for either Flock's employees or potential customers. Some examples of trainings we recommend include:

- Publishing a comprehensive training curriculum which goes beyond product use and covers secure configurations, implications of data retention, and goes over common privacy dilemmas that may show up while using their systems. This would help to increase customer understanding of how and for what Flock's systems can be used and reduce the risk of privacy leaks coming from misuse on the customer's side.
- Require structured ethical training on the impacts of civil liberties including case studies and criticism from groups like the EFF. Train executives and public facing personnel to respond to these criticisms with evidence based discussion rather than adversarial framing. Also train account managers and officials to conduct open forums and publish FAQs around sensitive privacy concerns. This would help to negate the framing of R15 and R17 and encourage more public discussion around privacy and ethics.
- Georgia tech should implement a more stringent signage policy which goes beyond the legal requirement of "area under surveillance". These signs should either outright acknowledge the area uses ALPRs or have a way for individuals to find their transparency page for Flock devices. A similar video surveillance notice is used by Southern Illinois

University [70]. Encouraging more notice will allow for individuals to understand how data is collected and gives them the opportunity to opt-out by maybe taking another route.

- Quarterly auditing to ensure privacy concerns are considered alongside any possible changes to the system. This would help to ensure long term compliance and would allow for Flock to monitor misuse and effectiveness as their company grows in size and scope.

Conclusion

In summary, we have conducted a privacy impact assessment on Flock Safety and their deployment in Universities. The following table summarizes the risks found in this assessment, along with our proposed mitigations.

Risk Level	Risk ID & Name	Proposed Mitigations
High	R2. Flock Vehicle Fingerprint	Allow for customizable amounts of vehicle data to be stored and default to the minimum amount
High	R3. Flock FreeForm Search	Enforce stricter data moderation policies
High	R5. Data Ingestion Pipeline	Ensure any third party data processors comply with the same privacy standards as Flock if not higher
High	R6. Training Data	Allow the customer to opt-out of having their data used for ML training
High	R7. Cybersecurity Vulnerabilities	Conduct frequent and high intensity audits to identify vulnerabilities
High	R8. Flock Safety Response to Security Concerns	Encourage transparency regarding cybersecurity vulnerabilities
High	R9. Flock APIs & Integrations	Ensure any third party data processors comply with the same privacy standards as Flock if not higher
High	R10. Drone Alpha	Pilots of drones must fly low enough or have cameras off as to not capture any information until at the scene
High	R12. Raven Detection Expansion	Require strong reasons for any possible

		expansions and only do so if absolutely necessary
High	R17. Camera Design and Installation	Enforce mandatory signage around areas covered by Flock's systems
Medium	R1. Data Storage & Retention	Enforce shorter data retention limits
Medium	R14. Software Expansion (Flock Nova)	Require strong reasons for any possible expansions and only do so if absolutely necessary
Medium	R15. Security-Centric Framing	Create a company policy of transparency and accepting of critiques
Medium	R16. Illusionary Control	Update the Safe List site and forms to give residents more awareness of exactly what Safe List does
Medium	R18. High-Friction Sharing Architecture	Make changing the sharing settings harder than a checkbox
Low	R4. Raven Audio Detection	Enforce stricter data moderation policies
Low	R11. LPR Expansion	Ensure the field of view of cameras is only pointed at active roads
Low	R13. Drone Expansion	Require strong reasons for any possible expansions and only do so if absolutely necessary
Low	R19. Chilling Effect on Activism	Enforce stronger requirements on search terms to prevent overuse

Figure 24. List of identified privacy risks and potential mitigations

Overall, the main purpose for deployment of Flock Safety onto university campuses is to keep students safe. However, universities already have existing infrastructure in LPR and security cameras, but Flock Safety enables them to be more interconnected, with better technology, and fill the gaps in their current infrastructure. Ultimately, it seems like it is specifically up to the university in how they choose to audit their systems, integrate Flock Safety onto their campuses, and who they share their data with.

While the system successfully achieves the goal of modernizing campus security, it does so by use of data collection, aggressive machine learning pipelines, and other surveillance designs. Our privacy impact assessment outlines the risks associated with the deployment of Flock Safety.

Appendix

GTPD Drone Pilot & Interview Request Summarize

From: McNevin, Richard F <richard.mcnevin@police.gatech.edu>
Sent: Wednesday, February 18, 2026 2:29 PM
To: Son, Andrew <ason31@gatech.edu>
Subject: RE: GTPD Drone Pilot & Interview Request

Hi Andrew,

I spoke to the Chief, and he approved me assisting you. Send me a couple of dates and times that work for you and we can meet to discuss.

Thanks,

Captain Rich McNevin

Office of Information Technology

Georgia Tech Police Department

965 Hemphill Avenue Northwest

Atlanta, Georgia 30332

Cell: 1-404-402-9078

Email: richard.mcnevin@police.gatech.edu

Learn: www.police.gatech.edu

Interview Request Georgia Tech Students CS 4726 Summarize

PL Paris Lewbel<paris.lewbel@flocksafety.com>
To: Son, Andrew

Reply Reply all Forward ...
Sat 2/21/2026 3:41 PM

This sender paris.lewbel@flocksafety.com is from outside your organization. Block sender Show blocked content

You don't often get email from paris.lewbel@flocksafety.com. Learn why this is important

Hi Andrew,

Thanks for reaching out. We respectfully decline your request for an interview.

Paris Lewbel
he/him
Public Relations Manager - Public Sector

icon 248.935.4712 icon flocksafety.com

App Banner Image

__tpx__

On Sat, Feb 21, 2026 at 12:59 PM 'Son, Andrew' via Media <media@flocksafety.com> wrote:

> To whom it may concern,
>
> My name is Andrew Son, and I am part of a team in CS 4726 class at Georgia Tech currently conducting a Privacy Impact Assessment (PIA) of Flock Safety. We would greatly appreciate the opportunity to interview Flock to better understand the ecosystem for approximately 15 minutes at your convenience. In addition, we would be happy to share the results of our assessment upon completion.
>
> Thank you for your consideration. Looking forward to hearing back from you.
>
> Sincerely,
> Andrew Son

Reply Forward

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